PackScan Bharat – Additional Proposed Features

These are proposed enhancement features and should be presented as proposed features unless they are already implemented in the project.

No.	Feature	What it does	Real-world example
1	Label Completeness Score	Gives the package a score based on how many required declarations are present and correctly detected.	An inspector scans a biscuit packet. MRP, net quantity and manufacturer details are found, but one required declaration is missing. The system highlights the missing information and gives a completeness score.
2	Smart Image Quality Check	Checks whether the uploaded package image is clear enough for reliable OCR before starting inspection.	Glare covers the MRP in a photograph. The system asks the inspector to capture the package again instead of producing an unreliable result.
3	Cross-Field Consistency Check	Compares related information on the package to detect contradictions or suspicious combinations.	A package shows different quantity information in two places. The system flags the mismatch for officer verification.
4	Unit & Format Validator	Detects unusual or inconsistent formats in values such as weight, volume, dates and MRP.	A product displays its quantity in an unusual format. The system identifies it and flags it for verification against the applicable rule.
5	Smart Violation Explanation	Explains what was detected and why it needs attention instead of only showing 'Violation'.	Instead of simply showing 'Non-Compliant', the system explains that a required declaration could not be detected and asks the officer to verify it.
6	Inspection Risk Score	Assigns a priority level based on the number and seriousness of detected issues.	Out of 20 inspected products, 3 have multiple potential issues. The system marks those inspections as High Priority for review.
7	Scan Again Comparison	Allows an inspector to rescan the same product and compare the new result with the previous scan.	An officer rescans a product after a retailer replaces the package. The system shows which detected fields changed.
8	Rule Simulation Mode	Allows an officer to test how a package would be evaluated under a different rule or rule version without changing the actual inspection record.	An officer wants to understand the effect of an updated requirement. The system runs a separate simulation and displays the expected result.
9	OCR Correction Assistant	Allows an officer to correct an incorrectly recognized word or value while preserving the original OCR result.	OCR misreads a manufacturer name because of decorative text. The officer corrects it while the original OCR output remains available as evidence.
10	Inspection Summary Card	Creates a quick summary containing product details, inspection date, detected declarations, violations and officer action.	After inspecting a product, the officer can view the important results in one summary card instead of checking multiple screens.
11	Product Category Detection	Automatically identifies the likely category of the packaged commodity before applying relevant checks.	The uploaded image appears to be a food product. The system identifies the category and prepares the relevant inspection checks.
12	Uncertain Result Flag	Separates confident results from results that need human verification.	OCR detects an MRP but confidence is low because the text is blurred. The system marks the result as 'Needs Verification' rather than treating it as final.